

Shaft Sinking

Raising, Boring Method

26/05/2019

Scope I

- *Development of raise boring & raise climber has opened up opportunities for shaft excavation by raising.
- *Method of shaft excavation is possible only when lower level has been development from another shaft, decline or winze.
- *Raising is recommended when there is no possibilities of sinking due to lack of free space in existing shaft & lack of room on production level for production purposes.

Scope II

- *This method can be utilized on in competent rocks of high & medium strength, with out fractures & with out a large water inflow.
- *Advantages of raising are listed below.
- *Raising is independent of mining production, hence mine performance is not effected.
- *Loading of muck into bucket is avoided.
- *Blasting effectiveness is greater.
- *No water pumping is required.

Scope III

It has following disadvantages

- *Height of raising is limited to about 100 to 120m.
- *Ladder climbing is inconvenience for personnel & transport of material to face.
- *There is potential hazard of pieces of rock falling from face.
- *There is potential threat of jamming in chute compartment (manual compartment raise).
- *Survey is difficult (deviation is possible)
- *There is time & cost involved in preparing opening for raising process.

Manual Raising I

- *Manual raising has three compartments, chute, hoist & ladder.
- *Brattice is installed from bottom of ladder compartment up to safety platform placed below working platform.
- *Chute compartment is of solid construction & is lined tightly with 50mm (2inch) planks (lagging) to withstand high side pressure of muck.
- *For long raise, a rebound niche is made every 30-50m in side wall of raise.

Manual Raising II

- *Working platform in raise are made with timber planks strong enough to with stand load of piled on muck.
- *This platform is supported by steel guarders or wooden beams with strength adequate to support wooden platform, weight of muck, equipment & workers.
- *For carrying steel beam, a safety factor of six is recommended.
- *Safety platform are lighter construction, lined with 50mm (2inch) lagging laid on beam of adequate strength.

Manual Raising III

- *There are two entrance with hatches, one in ladder & one in hoist compartment.
- *Distance between working & safety platform can not be less than 1.5m or more than 3m.
- *Chute compartment should be filled with rock when it has rebound platform in niches.
- *Only excess muck is removed by opening bottom damper.
- *In hard rock, a raise face can be advanced up to 5m with out lining & in weaker rock proportionally less.

Manual Raising IV

- *When lining consists of timber set, every third set has to be inserted in shaft wall.
- *When full timber lining is used, curb set support at least each platform of ladder compartment is done.
- *Chute compartment is guarded by a railing to prevent accidental fall of personnel & larger piece of rock.
- *For ventilation, a hole 200 to 300mm in diameter is drilled at centre of future shaft.
- *Hole is cased with in limit of sump of existing shaft & projects above water level in sump.

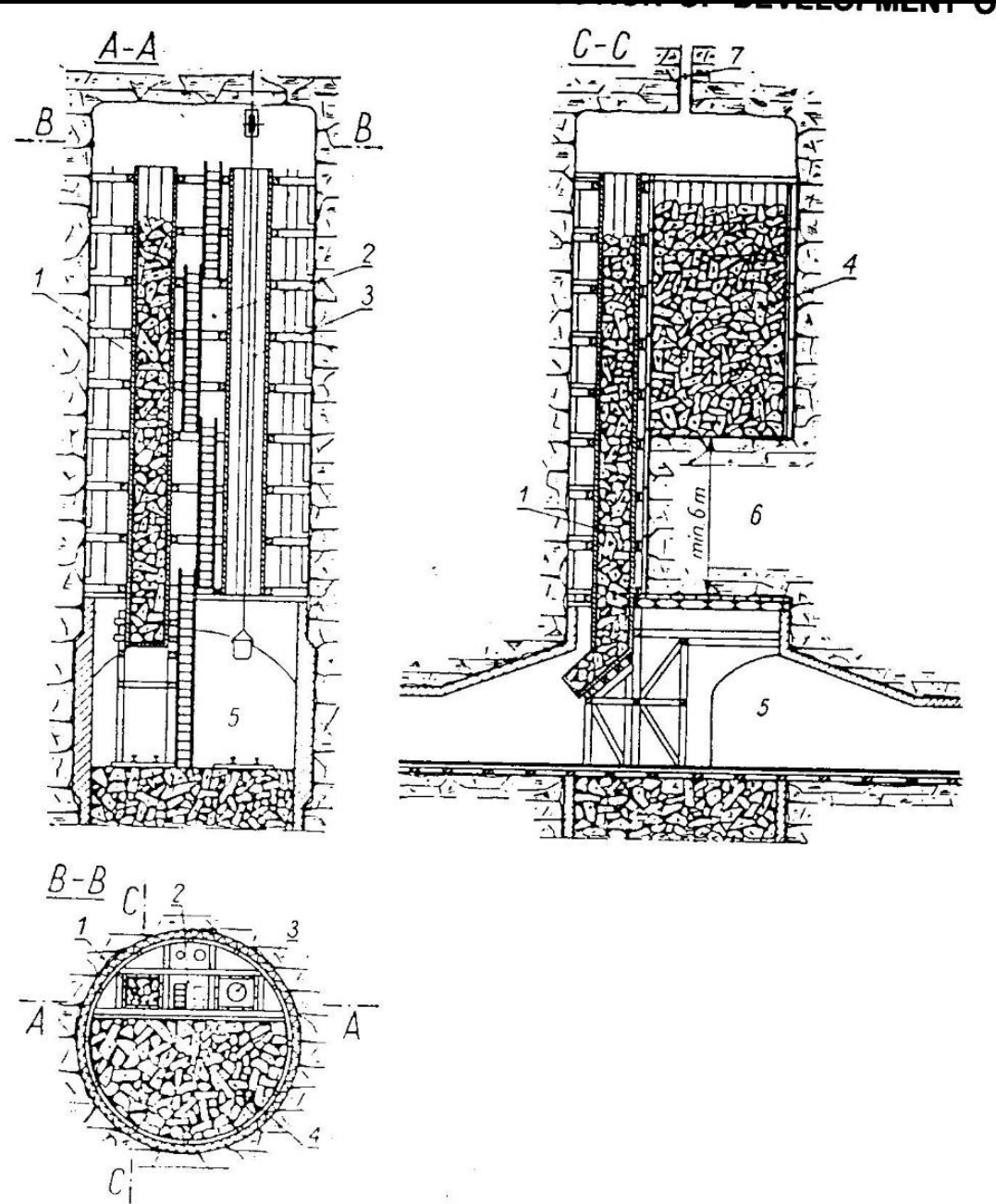
Manual Raising V

- *Three raising methods are used depending on size of shaft & geo-mechanical rock properties.
- *Raising shaft of small diameter, up to 4m, is done at full diameter with temporary lining.
- *A permanent lining is erected latter.
- *Good rock conditions are required.
- *In competent rock with medium strength small diameter shaft is raised full diameter with immediate construction of permanent lining.

Manual Raising VI

- * Shaft of diameter greater than 4 meter in competent rock of high & medium strength, first raise up ward at reduced diameter than small shaft is widened down ward to required size.
- *This is most common method of shaft excavation.

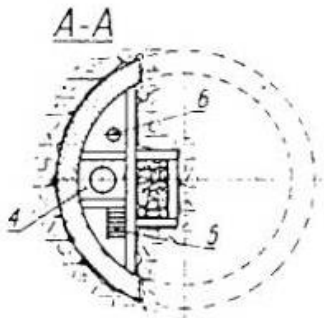
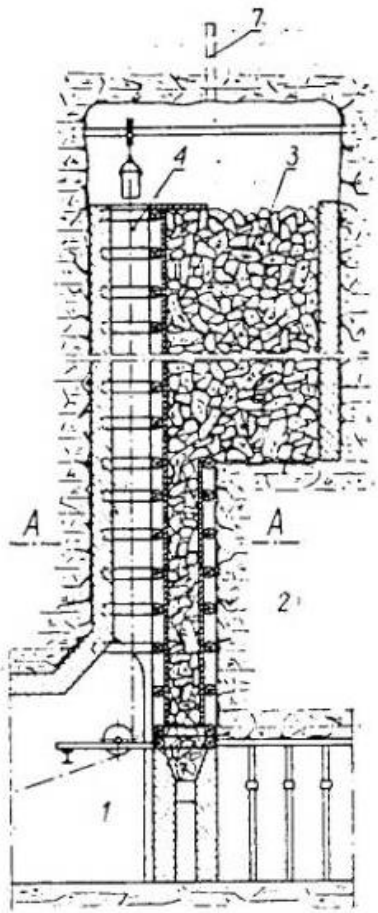
Manual Compartment Raising



Raising method with full diameter of shaft, with temporary & latter erected permanent lining.

- 1- chute compartment
- 2- ladder compartment
- 3- bucket compartment
- 4- storage of muck
- 5- lower shaft station
- 6- protective self
- 7- ventilation hole (to sump of existing upper shaft)

Manual Compartment Raising



Raising method with full diameter of shaft with immediate permanent lining.

1- shaft station on a new level

2- protective rock shelf

3- storage of muck

4- bucket compartment

5- ladder compartment

6- tubing compartment

7- drill hole for ventilation (to sump of existing upper shaft)

Shaft Raise Climber I

- *Raise climber runs on guide rail that incorporate a pin socket.
- *Segment guide rail is fastened to side of excavation with rock bolts.
- *Alimak method has following advantages over conventional system.
- *Permitting excavation of very long raises, vertical or incline, straight or curved.
- *When travelling to head of raise, persons are well protected in a cage under platform.

Shaft Raise Climber II

- *Man work from platform that is easily adjusted for convenient height & angle.
- *Risk involving gases are reduced, raise is ventilated after blasting with air-water mixture supplied by raise climber.
- * All equipment & material required can be taken to head of raise in raise climber.
- *Timber is avoided, if needed, rock bolts, with or without screen are used for support.
- *This raise climber can be used for over up to 8sqm with single platform.

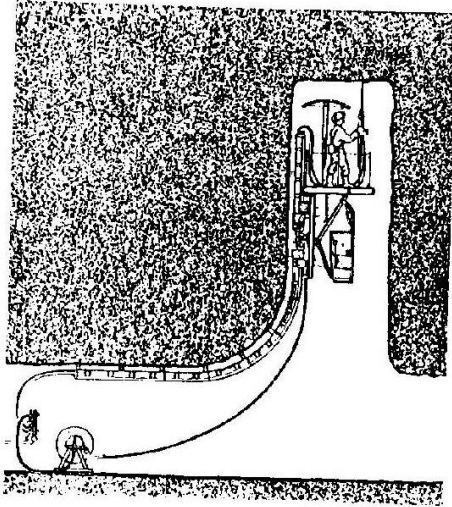
Shaft Raise Climber III

Disadvantages

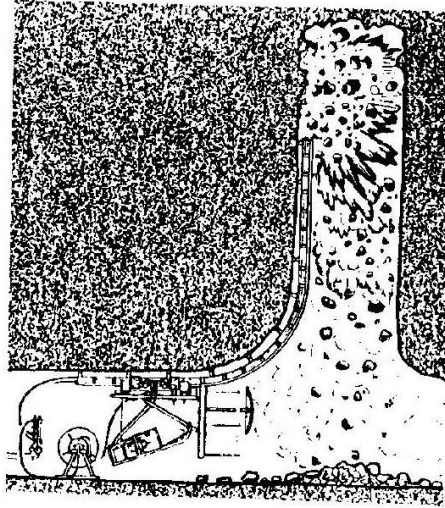
- *Raise climber is not suitable for short raises.
- *Present days raise climber are using electric, electro diesel power to excavate long raises.

Raise Climber

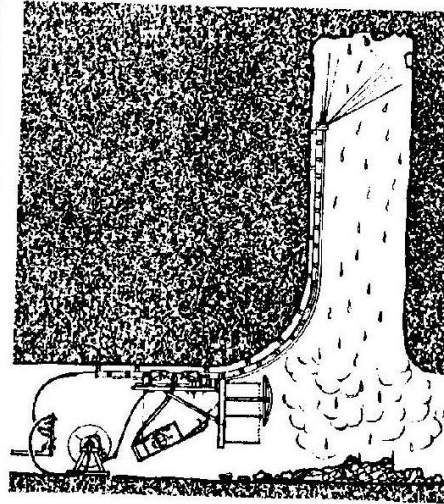
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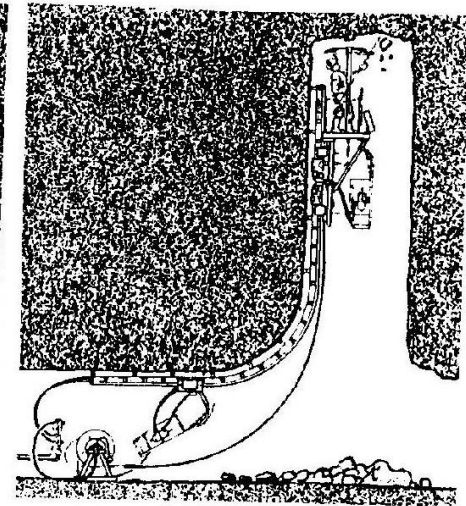
DRILLING: Scaling and extension of guide rail is safely and easily carried out from the platform under the steel protective roof. Tubes in the guide rail are used for the supply of air and water to the drills.



BLASTING: During blasting the Raise Climber is well protected in the tunnel.



CLEANING: After blasting the face is cleaned with an air and water mixture sprayed through jets in a plate protecting the top of the guide rail.



CLIMBING: The air motors are controlled from the cage, and they are supplied with air via a hose connected to a reeling drum in the tunnel. ALICAB or ALITROLLEY Reserve and Service Lifts provide fast communication with the Raise Climber and is useful for the transport of men and material, or in the case of accidents.

Shaft Boring I

- *Large diameter holes or shaft can be drilled vertically or inclined.

Shaft borer has following advantages

- *No worker need to be present at shaft face.
- *This reduces exposure of men to hazards condition.
- *Reducing man power increases cost effectiveness (cost reduction).
- *Continuous boring assures faster completion of project.
- *Boring does not disturb surrounding rocks.

Shaft Boring II

- *There is no need of support or substantial reduction in support requirement.
- *Perfectly rounded shape makes excavation inherently stable.
- *Its smooth wall reduces resistance for ventilation as well as for passing ore.

It has following disadvantages

- *General requirement of rock condition be similar with in length of raise bore.
- *Drilling in hard rock is slow & cost rapidly increases.

Shaft Boring III

- *Access to upper & lower levels is necessary.
- *Raise bores rock has to be dry.
- *Initial cost of rig is very high.

Shaft Boring Equipment I

- *Raise boring are originally designed for soft rock.
- *Now machines are available for hard rock.
- *Different raise boring machines are available for different rock conditions, nominal sizes & length of finished hole.

Machine specification includes following

- *Pilot hole diameter.
- *Drill pipe diameter
- *Reaming torque
- *Type of drive (AC, DC, hydraulic)
- *RPM

Shaft Boring Equipment II

- *Power kw
- *Type gear reduction & RPM
- *Pilot hole thrust
- *Feed rate
- *Reaming pull
- *Size of drill pipe
- *Base plate.
- *Derrick dimension with characteristic
- *Type of transport / erector
- *Water consumption
- *Air consumption

Shaft Boring Equipment III

- *Raise boring machines operate on principle of first drilling a small pilot hole & then reaming hole in one or more stages, to achieve desired size.
- *Normally pilot hole is drilled down & reaming hole up wards.
- *Some time single stage drilling of raise is also adopted.
- *Rig is itself a rigid plate & structure, on which drilling, hydraulic equipment are assembled.
- *Rig has its own crawler driven by compressed air or on rail wheel for haulage.

Shaft Boring Equipment IV

- *Rig can be dismantled & hauled in underground service type vehicle.
- *Pilot hole is drilled through stem with stabilizer & conventional drill bit.
- *Drill rod has outer / inner diameter in range of 20.3 to 30.4mm (8 to 12.7/8 inch) to 12.06 mm to 13.1mm (4.3/4 to 5.7/16 inch).
- *Length of section for underground application is usually 1.5m, dictated by clearance restriction.
- *Connection are made with tapered thread joints that have to be kept absolutely cleaned.

Shaft Boring Equipment V

- *Purpose of stabilizer is to assure directional accuracy of pilot hole.
- *They are larger diameter than drill rods, with strap of tungsten carbide welded vertically or spirally on outside, bringing outer dimension close to hole diameter.
- *These stabilizers are used for reaming up.
- *Stabilizer for pilot holes & reaming down, have more clearance, allowing cutting to pass.
- *Once pilot hole penetrates to mine opening, pilot bit & stabilizers removed.
- *After necessary servicing, stabilizer is mounted on drill stem & reamer is placed back.

Shaft Boring Equipment VI

- *Starting hole (collaring) is a critical operation.
- *Damage to cutter / bit body can occur due to uneven surface.
- *Tension on stem has to be adjusted to allow starting hole with out hitting variable dynamic load acting on drilling component.
- *Stem is longest at this stage of drilling & there fore, prone to maximum elongation & twist.
- *Design of reaming depends on final size of hole & number of reaming stage.

Shaft Boring Equipment VII

- *Standard analysis is required & is usually performed by computer, taking in to account necessary balancing tools, achieved by optimization of cutter.
- *Some time, catenation are attached to primary body of cutter.
- *By this, an increase of final diameter is achieved by addicting more cutter in same plane as cutters on primary body itself.

Boring To Shaft Diameter I

- *When full size shaft diameter is to be reamed, primary body of reaming is supplemented with stage mounted below.
- *General gears are being used for reduction of required torque, when compared to conventional reamers.
- *Bit turn at slower rate than drill pipe due to gearing mechanism.
- *Shaft reamers are used for very diameter holes.
- *This is a separate machine, that is installed after standard size raise has been completed.

Boring To Shaft Diameter II

- *It is used for enlarging raise to full diameter required,
- *Shaft reamer stabilizer it self by side jack against drill hole wall.
- *This develops against anchored portion of machines.
both drilling load & torque reaching
- *Cuttings fall freely down through previously completed raise.
- *Shaft lining can be installed when drilling shaft.
- *A service hoist to working level is installed for men & material.

Raise Boring To Shaft Diameter III

- *Drilling performance is measured by rate of penetration.
- *Drilling performance for a particular rock greatly depends on bit design.
- *Cutting action has to match rock penetration to optimize drilling rate.
- *Design of reaming head influence raise boring process while reaming.
- *An analysis of presently used cutter can help in design improvement of cutting standard.

Boring Equipment I

- *Independent variable characteristic of drilling performance are compressive, tensile & shear strength.
- *Drilling tool should break rock primarily in tension & shear, because strength of rock in these modes of loading are minimal.
- *Rock ductility & abrasibility also influence drillability of rock
- *Strong but brittle rock has better drilling response than weak, clay rock.

Raise Boring Equipment IA

- *Abrasibility depends on mineral content of rock & is responsible for wear of drilling tool element in contact with rock face & cutting.
- *Properties of rock mass, joints, faults, structured features also influence drill response.
- *Layer of cutting on uncut rock reduces penetration up to 40%.
- *Energy of drilling is spent on regrinding cutting rather than penetrating intact rock.
- *In raise boring, this situation is encountered during drilling of pilot hole.

Raise Boring Equipment BI

- *Drilling technology with compressed air, air & foam, drilling fluids is used for cutting removals.
- *During reaming operation, cutting falls down the hole.
- *Drilling load should be at level required by cutter to penetrate & create fracture.
- *Once threshold strength rock is reached, further increase in drilling load does greater depth penetration to local, but not in direct proportion.

Shaft Boring Equipment CI

$$D=Wd$$

D> depth of penetration

W> approved load

d> load exponent (1.5 in soft rock, 1 in hard rock)

*Operator measures rate of penetration over several minutes.

*He then adjust load through several trials to find optimum load, above which rate of penetration does not noticeably increases.

*Rotary speed also effects penetration rate.

*Higher rotary speed, faster rate of penetration.

Shaft Boring Equipment I

*Other restriction in drill unit are wear of bearing & drilling tools, vibration on uneven surface, rubbing of drill pipes against drill hole

Shaft Boring Method Selection I

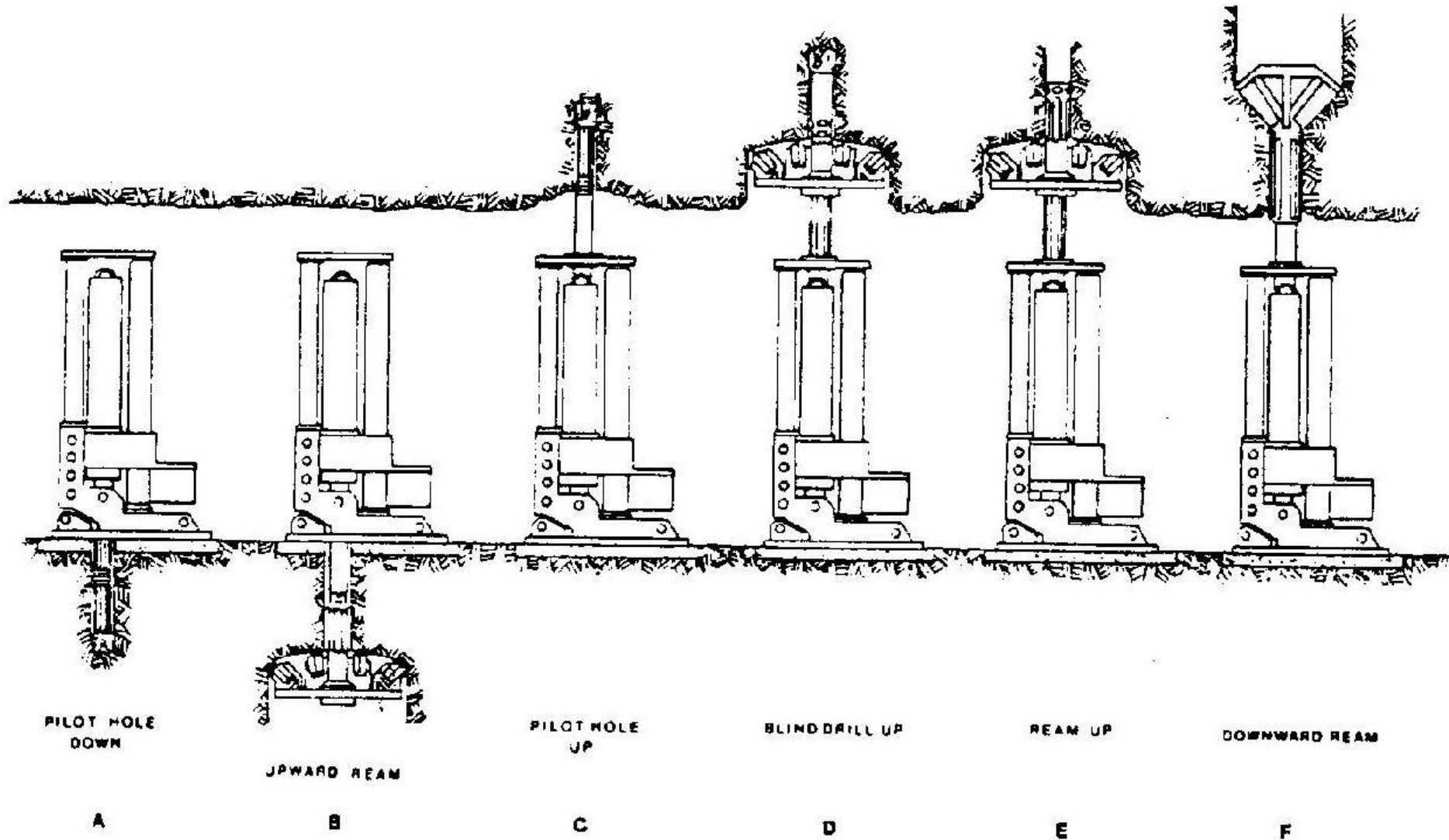
- *Decision whether to drill raise from top to bottom or bottom to top usually determined by access point.
- *If only one end of raise has adequate rooms, machine must be put there.
- *Location of borer is placed such a way to create least disturbance in mine.
- *Generally, drilling down & reaming up is followed, because this is cheapest method.
- *Rock quality & its predicted response to drilling can influence choice of direction.

Shaft Lining I

- *In strong rock without fracture usually no lining is required.
- *If support is required, shotcrete, rock bolts, with or without wire mesh or steel lining can be applied.
- *Shotcrete can be placed remotely or by worker from cage.
- *Rock bolts are to be installed manually.
- *Steel lining, required in unstable formation, is installed remotely, directly behind reamer.

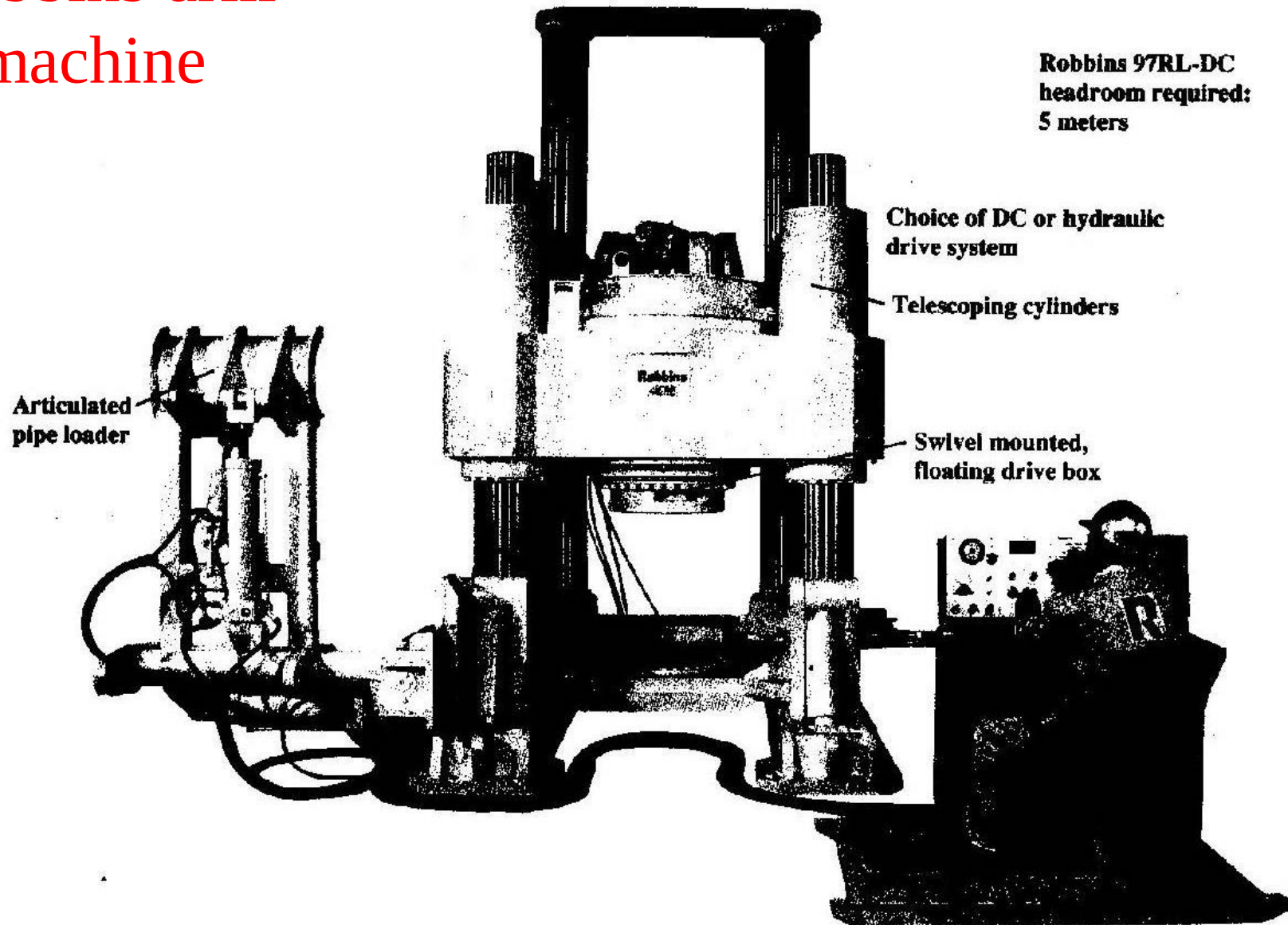
Shaft Boring

Modes of operation in shaft boring



Shaft Boring

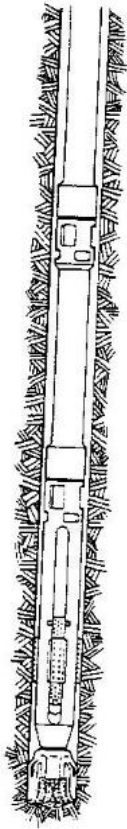
Robbins drill machine



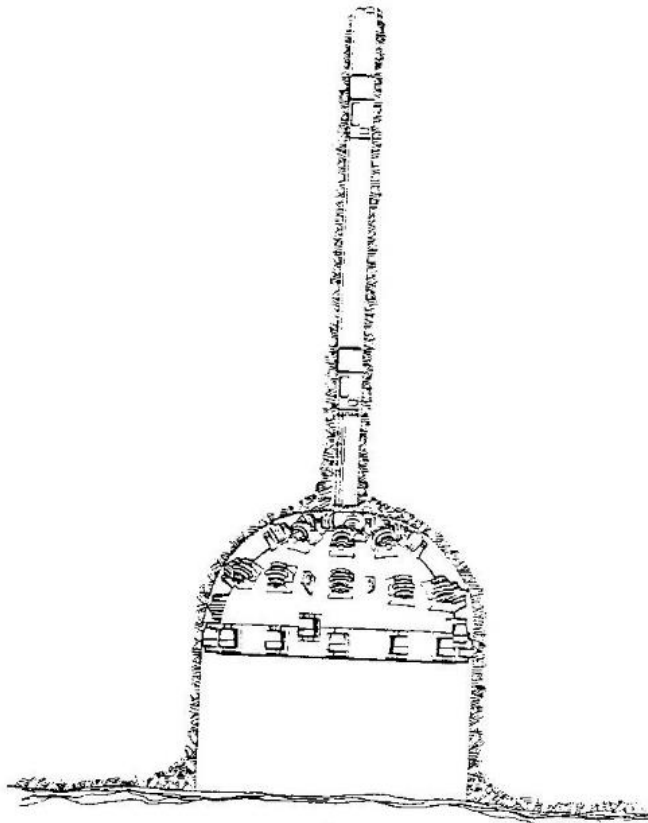
Shaft Boring

CONSTRUCTION OF

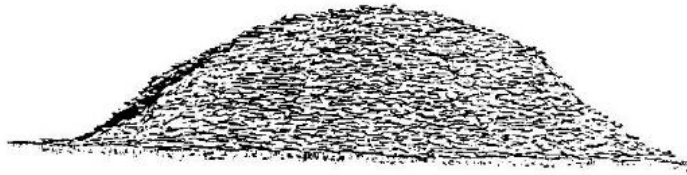
Pilot hole (a) drilling & (b)
reaming



(a)



(b)



Shaft Boring

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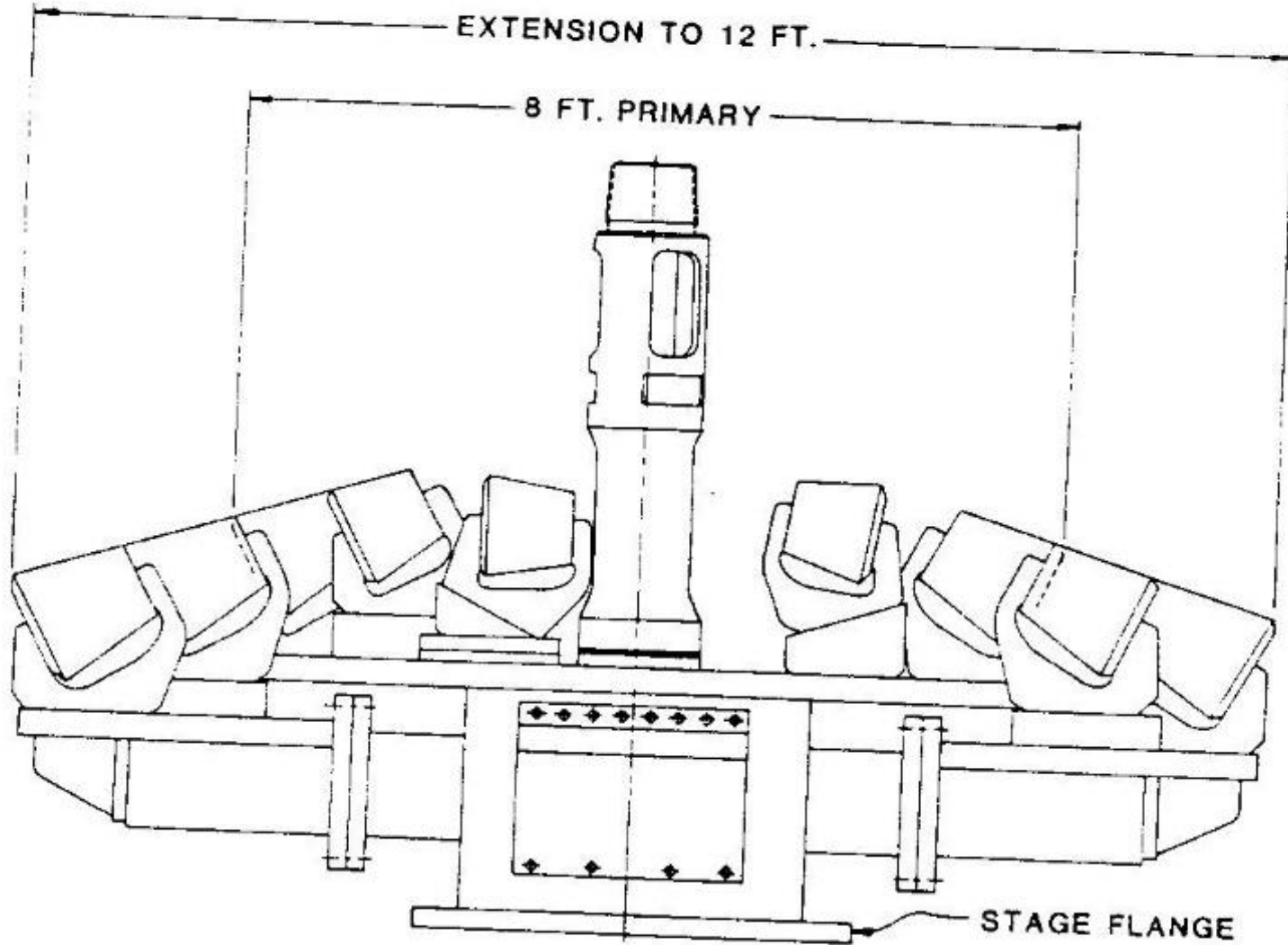


Fig. 17.4.64. Raise bit with extensions. Conversion factor:
 $1 \text{ ft} = 0.3048 \text{ m}.$

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Shaft Boring

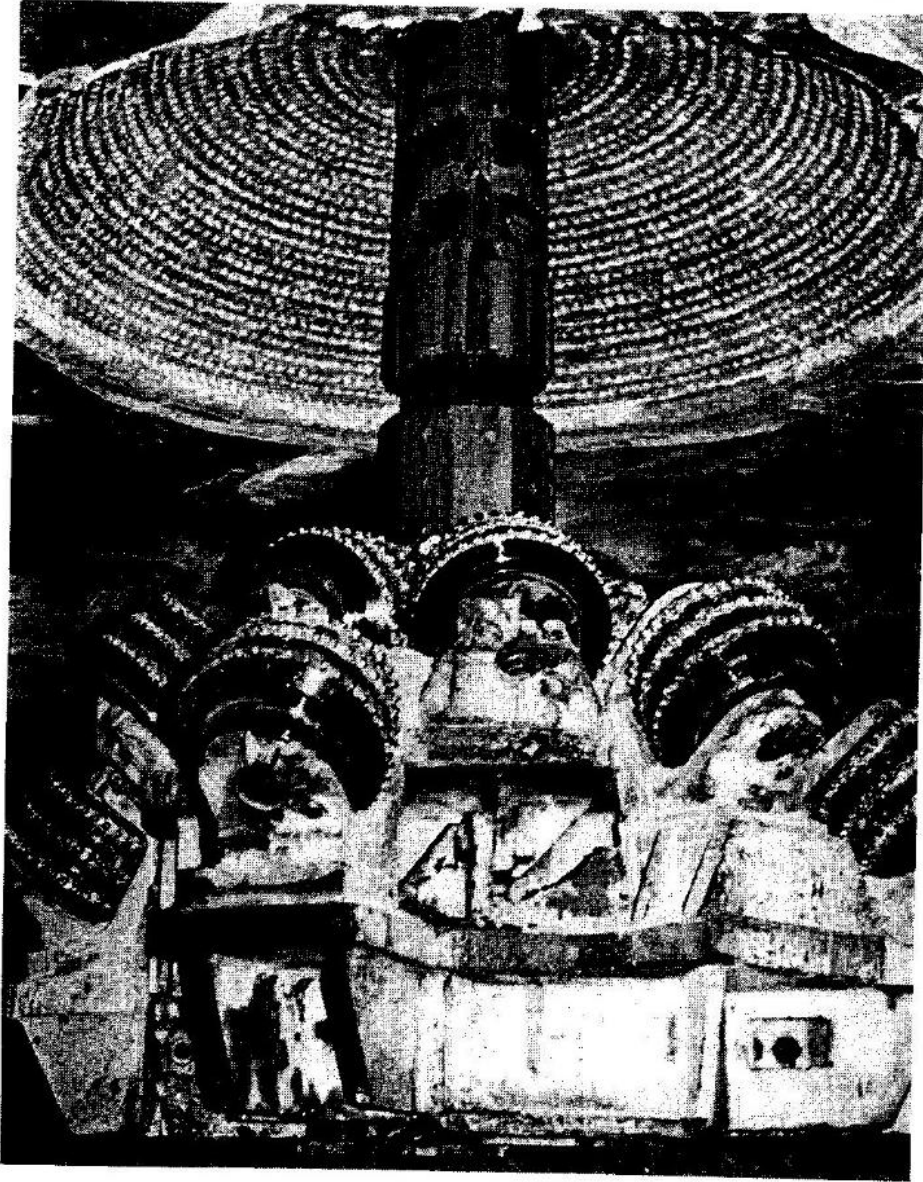


Fig. 17.4.65. Top section of 16-ft (4.8-m) diameter reamer breaking through the surface after 1490 ft (453 m) of reaming.

Shaft Drilling I

- *Shaft Boring is modern method for shaft development.
- *Shaft boring eliminates hazards placing of persons in sinking shaft.
- *Boring method is a continuous method (not cyclic like conventional method), hence sinking is faster.
- *In present technology, shaft up to 1000m can be bored with 10m diameter.
- *Rapid development in drilling technology, has enable to bore shaft up to much greater depth & larger diameter.

Shaft Drilling II

- *Shaft boring can be applied in difficult ground conditions.
- *Present shaft boring drill rig can drill in rock strength up to 300 Mpa.
- *Drilling system to be decided based on following
 - *Hydrological Conditions.
 - *Geotechnical parameters of rock.
 - *Shaft profile.
 - *Shaft Diameter.
 - *Shaft Depth.
 - *Type of cutter for rock to drill in.

Shaft Drilling III

- *Selection & technology of lining placement (buoyancy or casing type)
- *Determination of footing type.
- *Location.
- *Fastening & sealing water within aquifers, if present.
- *Organization of drilling site.
- *Location of pond for drilling.
- *Drilling fluids.
- *Site for fluid making plant.
- *Stage of lining component & materials.

Shaft Drilling Fluid I

- *Drilling fluid is a mixture of water & natural clay partly dispersed & partly colloidal.
- *By virtue of hydrostatic head, drilling fluid keeps wall of hole in an equilibrium state.
- *Drilling fluid transport chips of drill rock & cools bit.
- *Normally drilling fluid is prepared in mixture from which it flows to a tank & pumped by slurry pump to shaft.
- *During drilling, shaft has to be filled completely with drilling fluid.
- *To impose movement of fluid, a compressed air lift is used.

Shaft Drilling Fluid II

- *In smaller holes, where annular space between drill string & hole wall is small normal (right) circulation is applied.
- *Air is injected in this space & drill fluid flows to bottom drill pipe, raising around it to lift cutting to surface.
- *For large holes with a large sectional area of annular space between drilling string & wall, reverse system is used.
- *Air assisted, so called reverse (or left) circulation system can be applied as single, double & triple wall pipe.

Shaft Shaft Drilling Fluid III

- *In simplest case a 75mm (3 inch) pipe is run down drill string, approximately 100m, some time to end of string.
- *Compressed air is injected through 75mm pipe.
- *When it raises in annulus between inner & drill pipe it creates a mixture of air–drilling fluid of decreased density.
- *Positive pressure of drilling fluid (with unchanged density) in in shaft increases circulation with a resulting high flow velocity in drill pipe, enabled to transport of cutting to surface.

Shaft Drilling Fluid IV

- *More viscous fluid, less fluid velocity needed..
- *Usually, maximum particle load system can carry is 5% of its volume.
- *In recent development, a submersible pump is installed as a part of drill string.
- *This allows an increase in volume of circulating flow (in drill pipe 300mm bore, 15 to 20m³/min)
- *This enable better evacuation of chips from face.
- *To increase size of cuttings transported, in addition to enlarging flow velocities, cross sectional are with skirt & shrouds at very bottom of hole.

Shaft Drilling Fluid V

- *Rock chips shall not be lying exposed to unnecessary regrinding by bit.
- *Changing angle if sucking hole & adding chemicals to drill fluids can also serve this purpose.

Shaft Drilling Equipment I

- *Drilling rig for shaft boring must be able to produce large torque & required hoisting capacity must be large.
- *Verticality of large blind hole is a necessary condition that can be achieved by guiding bit by a pilot hole or by plumb bob effect of very heavy end of drilling string with large bit, weight & stabilizer
- *For very large diameter hole it may be necessary to use two or more passes with different size of bits.
- *Weight above drilling structure assures verticality of hole.

Shaft Drilling Equipment II

- *Weight produce load requirement for bit to penetrate rock.
- *Weight prevents reversal stress on drilling string by fly wheel effect.
- *Weight smoothens dynamic reaction on string from jerks & bumps caused by rolling over rock with unevenly distributed strength.
- *Stabilizer can be of two types rotating & non rotating.
- *Stabilizer rotate together with string & is often equipped with rollers.

Shaft Drilling Equipment III

- *These rollers some time have cutting structure & works as reamer.
- *Non rotating stabilizer have outer shell or pad that remains stationary, while inner core rotates.
- *These are appropriate for softer rock because they do not damage shaft wall formed by weak material.
- *Stabilizers assures vertical alignment of hole by restricting lateral movement of drill bit.
- *By using multiple stabilizer, it is possible to drill shaft with 1:1000 accuracy.

Shaft Drilling Equipment IV

- *Breaking of rock at base of shaft (hole) is achieved by cutters attached to bit body.
- *Durability & life of cutter is very important, permitting increased loading to be applied.
- *Milled steel tooth cutter are most suitable for soft rock, producing large cutting.
- *This gives potential for rapid rate of penetration, but performance of cutters diminishes rapidly as teeth becomes dull.
- *Tungsten carbide cutters are more appropriated for harder rock or in particular abrasive formation.

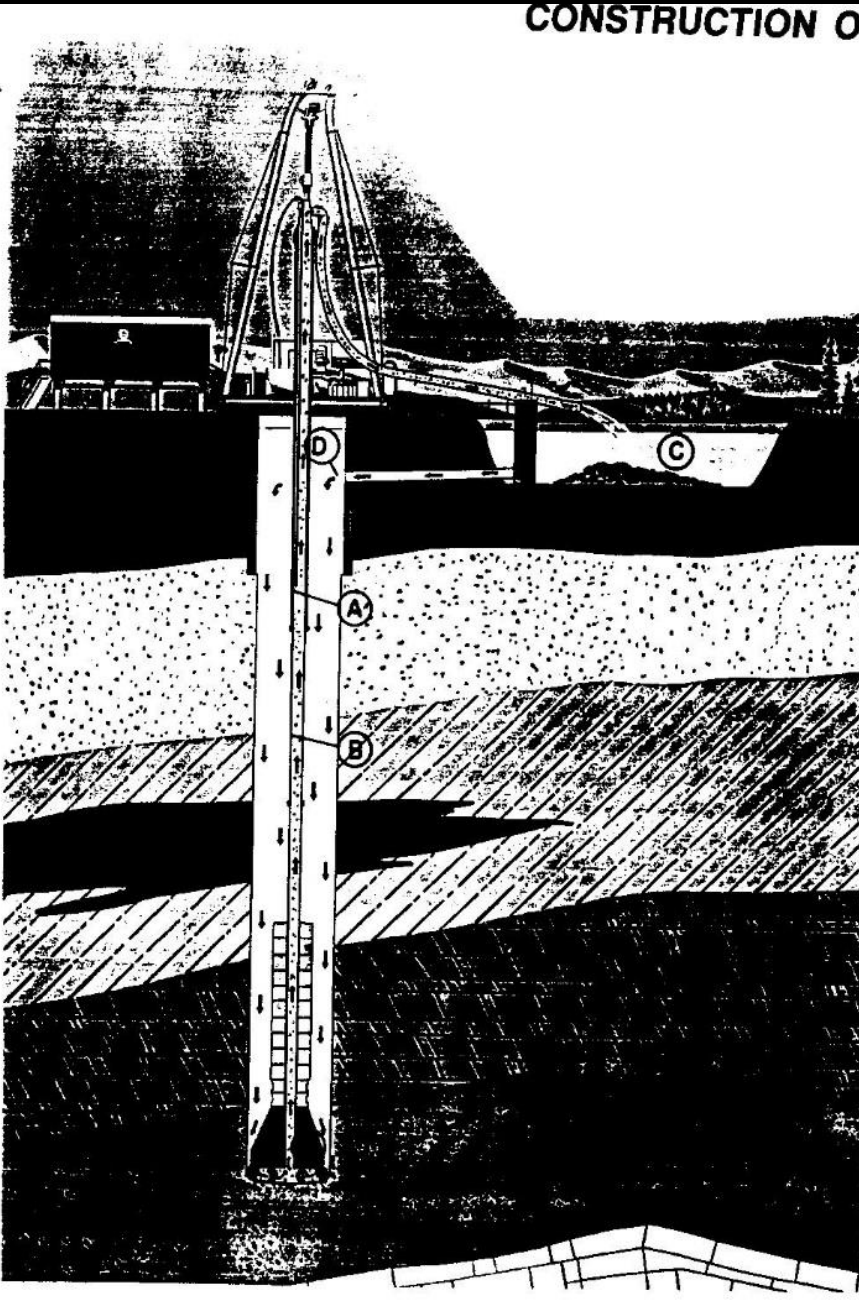
Shaft Drilling Equipment V

- *These bit requires substantially longer bit loading than other bit type & produce smaller chips.
- *Rate of penetration is slow, although performance remain steady through out life of cutter.
- *Disk or kerf cutter are not normally used for blind hole drilling because they tend to produce large chips that can not be transported across bottom of hole & up to surface.
- *In soft & plastic strata, they can produce concentric furrows in base of bore with no excessive progress.
- *Cutters themselves are mounted in saddle that are attached to a fabricated steel body of bit.

Shaft Drilling Equipment VI

- *Various basic shapes have been used, including flat bottom, conical, hemispherical or combination of shapes.
- *There is not enough evidence to determine which shape is most advantageous.
- *It is because of varying conditions in a particular project make such comparison difficult.
- *Some believes that flat bottom bits drills straight holes than conical bits.
- *Conical bits give better rate of penetration in medium to soft formation.

Shaft Drilling Equipment I



Blind shaft drilling

C- chamber to collect cutting fluid & drilled material.

D- cutting fluid to drill hole

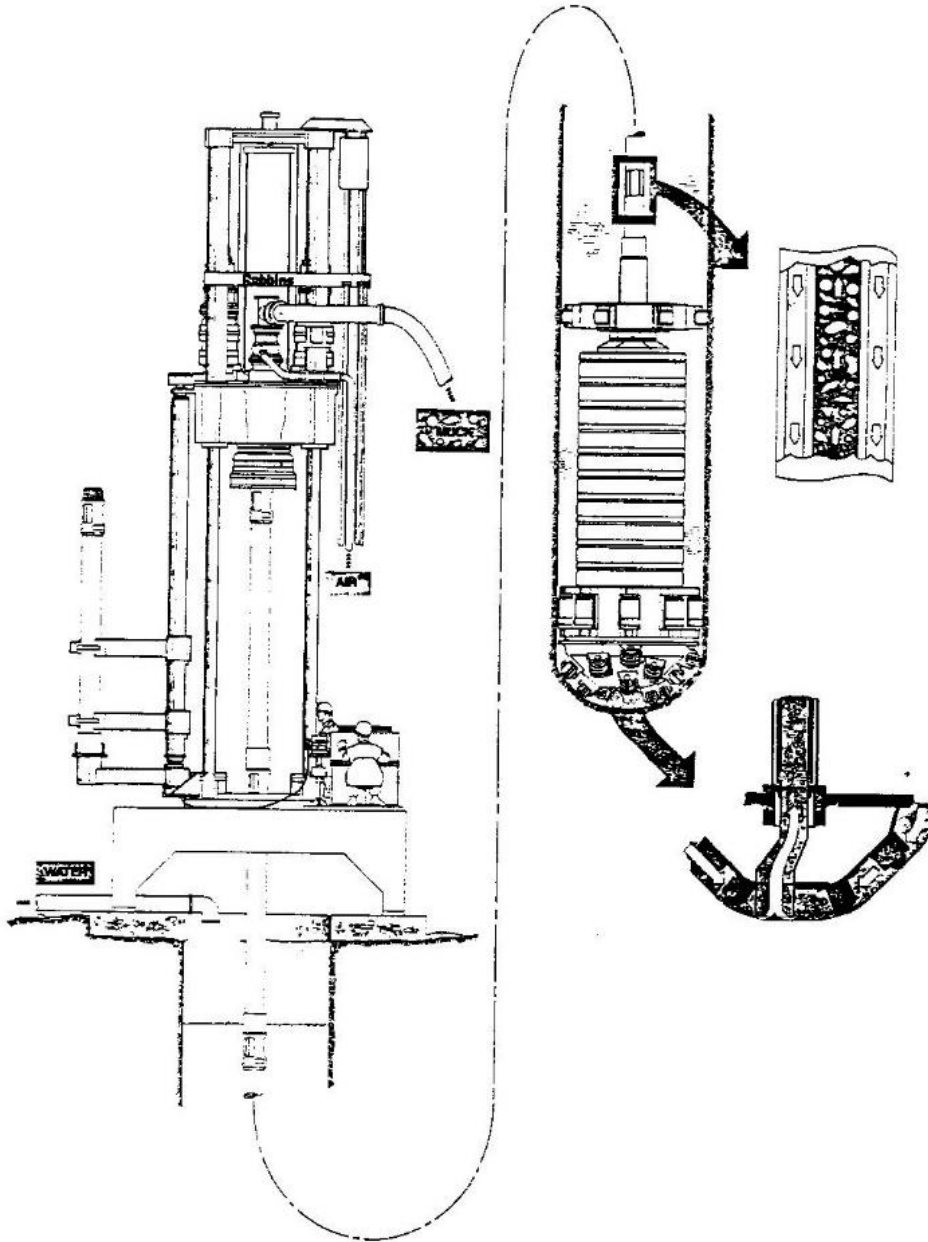
A- cutting fluid with cutting moves up through drill pipe.

Shaft Drilling Equipment I

EQUIPMENT OPENINGS

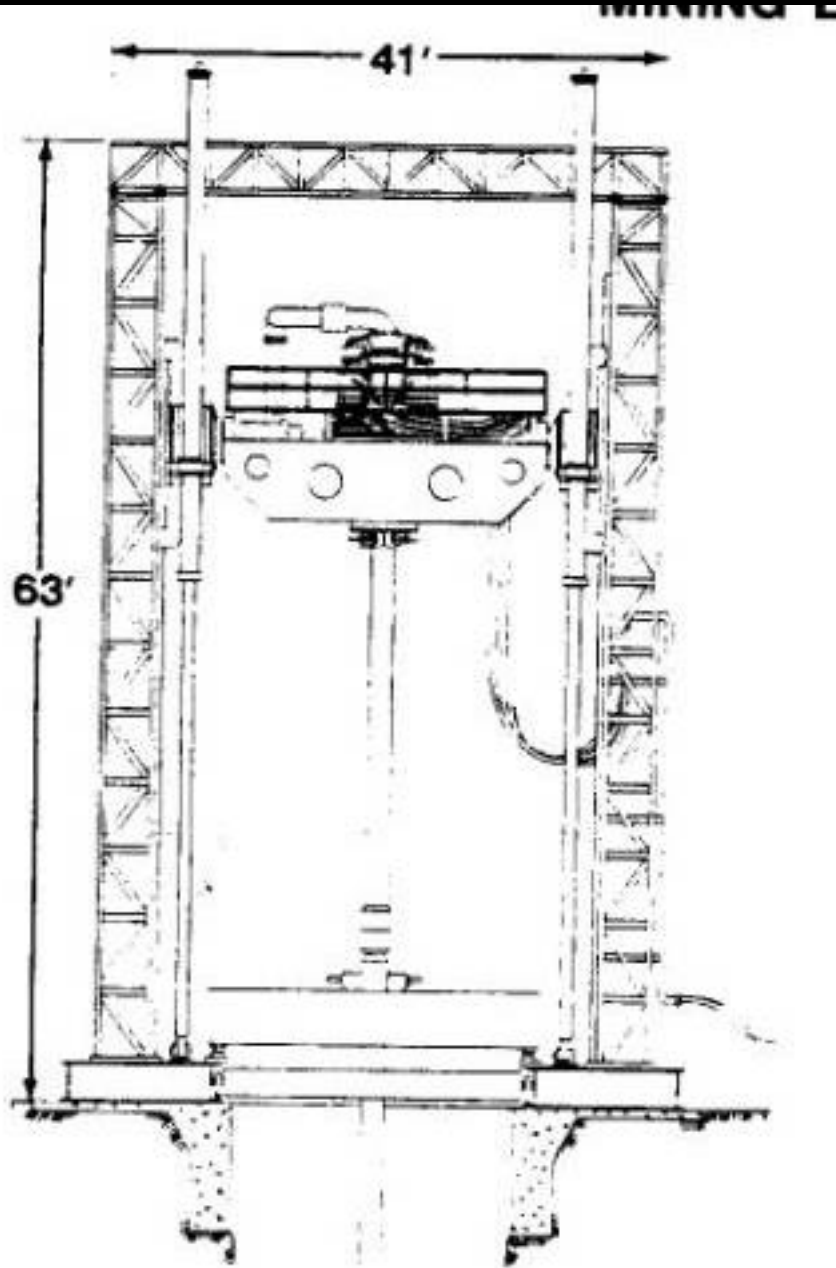
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*Blind Shaft Drilling
Rig, Robbins



Shaft Drilling Equipment I

Hydraulic Shaft Drilling Rig Hughes-Micon



* Shaft Drilling Equipment I

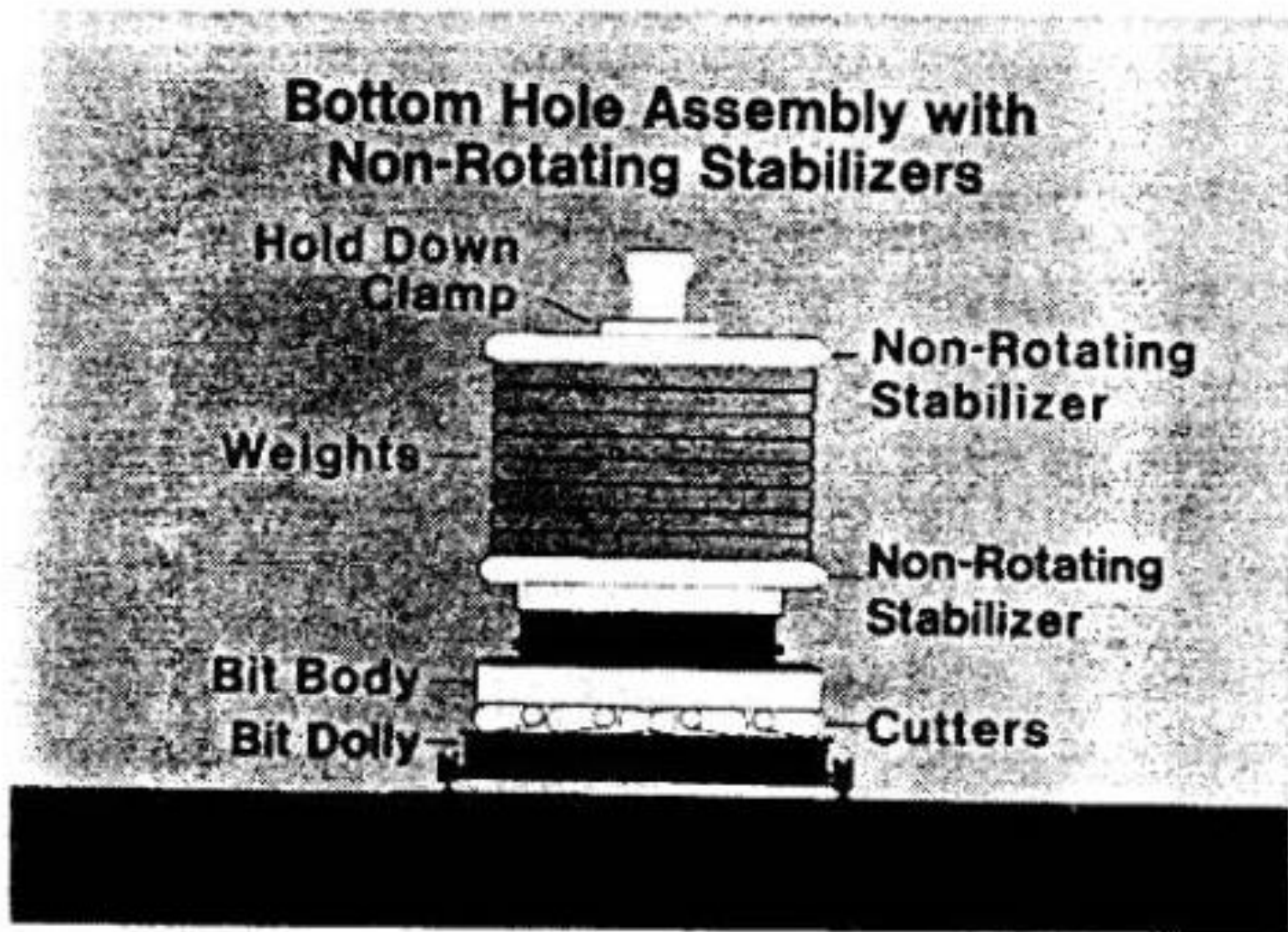


Fig. 17.4.71. 14-ft (4.3-m) diameter bottom-hole assembly.

Lining Shaft Drilling I

- *Type of permanent lining for drilled shaft depends on character of rock formation intersected by shaft.
- *In strong rocks, minimum lining requirement exist, hence shotcrete usually sufficient.
- *In fractured zone, a combination of rock bolts with mesh & shotcrete can be applied.
- *For shaft with hoisting installation usually a minimum thickness of 200mm of concrete lining is required for fastening of consoles for shaft beam.
- *This type of lining is installed from hanging stage using conventional technology.

Lining Shaft Drilling II

- *Permanent lining for water bearing, weak strata can be made of reinforced concrete or steel as a single or double cylinder, with concrete fill between outer & inner segment.
- *Work is performed at surface & lining is lowered to shaft filled with drilling fluid.
- *In floating method, a steel bottom is made (welded) beside shaft & set afloat in shaft.
- *On bottom, a first segment of lining. 2.5 to 3.5m high is welded.

Lining Shaft Drilling III

- *Existing lining with consecutive segments causes weight of lining to increase & consequently bottom sinks.
- *Drilling fluid is displaced by lining with equivalent volume.
- *When lining has positive bouncy, by pouring water in to it, required balance can be achieved.
- *Rate of lining placement is about 10m/day.
- *If required by depth of shaft & formation pressure, a double steel cylinder lining with concrete filling space between them can be made.

Lining Shaft Drilling IV

- *Steel segments pre fabricated are with required curvature & made by automatic welding machine.
- *Concrete inside lining is usually made with application of concrete method.
- *Injectors are placed vertically with uniform spacing.
- *Space is filled with gravel, sand, & aggregate (mixture in desired proportion)
- *Cement of high quality is pumped in to stone skeleton through injector, filling void from bottom up & pushing air out.

Lining Shaft Drilling V

- *Vibrators are used for up grading composition & strength of concrete (40 to 50 Mpa after 28 days).
- *If for some reasons, any difficulties appear during sinking operation, lining process can be turned on.
- *In case of problem with shaft wall, shaft can be filled with drilling fluid in stead of water to make fill heavier.
- *Some time reel drum or hydraulic hoist are used for lining shaft.
- *15 to 25 meter shaft length can be lined by hoist with out bottom.

Lining Shaft Drilling VI

- *This system is more appropriate for lighter single cylinder lining, sufficient for condition with lower formation pressure.
- *After lining, entire length of shaft annular space between lining & shaft wall is filled with cement clay grout (proportion 2:1 & specific gravity 1.65 to 1.75)
- *This grout is pumped upward through hoses from bottom.
- *Grout seals contact of particular lining section in shaft.

Lining Shaft Boring VII

*If required, additional sealing can be done for stages by grouting.

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SBM, Shaft Boring Machine I

- *Shaft boring is newest & probably latest mature technology of shaft sinking.
- *In shaft boring system, operators are travelling on board & muck transportation , services facility provided to surface.
- *Shaft boring machine consist of cutter wheel with a horizontal axis mounted on carriage.
- *Carriage is mounted on slewing structure, that rotates about vertical axis of shaft.
- *Entire assemble is position in shaft by an upper & lower gripper system that engages side shaft.

SBM, Shaft Boring Machine II

- *Walking cylinder between gripper assemblies allow machine to achieve up & down in shaft in increments.
- *A clamp type mucking machines unit is mounted opposite cutting wheel on slew structure & loads directly from shaft bottom in to hopper.
- *A conventional mine hoisting system allows muck bucket to be lowered in to SBM to transfer muck from hopper to surface.
- *Cutter wheel it self fitted with disc cutter & powered by two 261 kw motor.

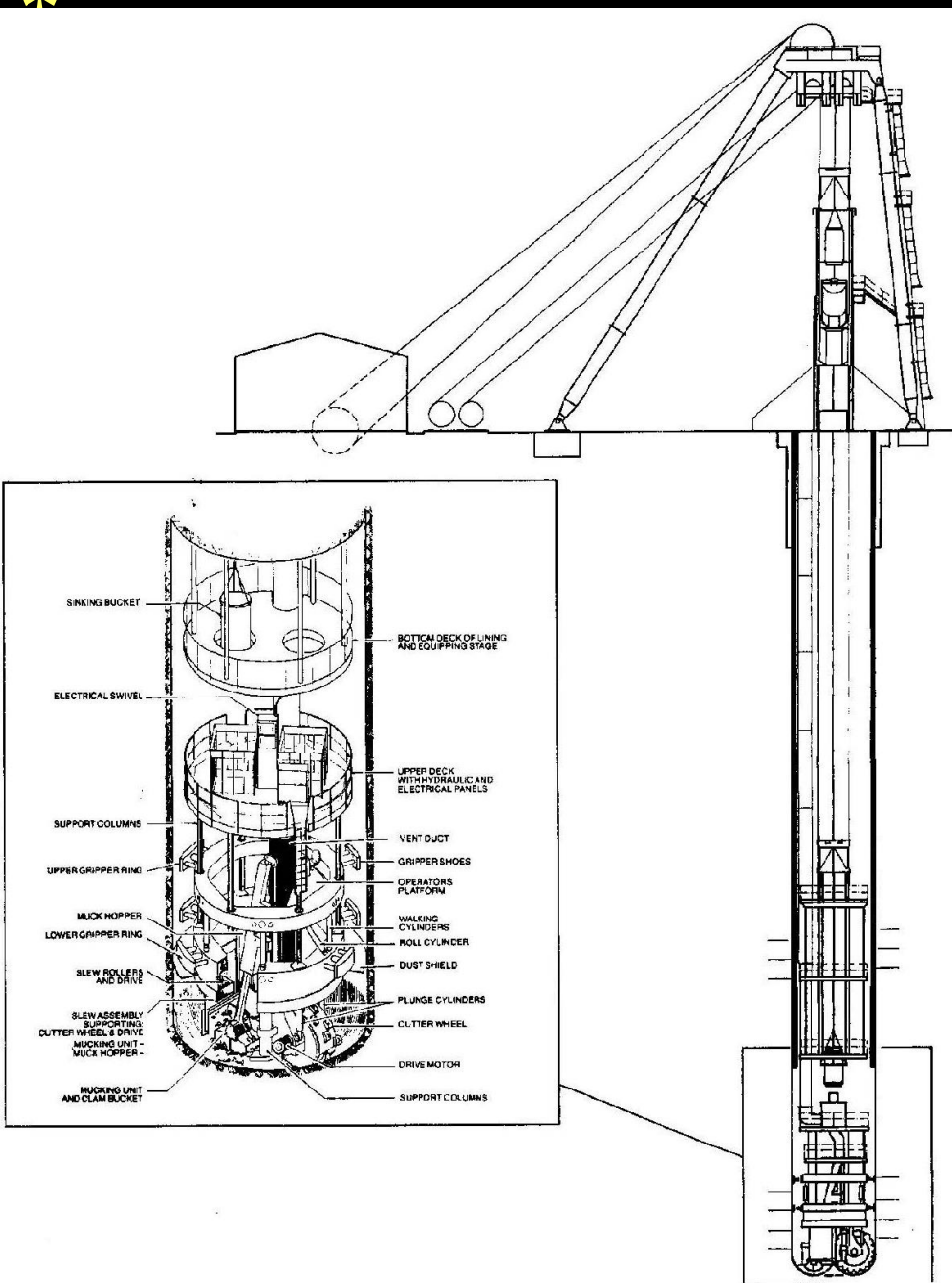
SBM, Shaft Boring Machine III

- *Disc cutter output is in lump form & it is easier for clamp type loader to load on bucket.
- *All component of machine fits through 2m X 2m opening, with longest component approximately 5.5m in length.
- *Total weight of SBM is about 14 ton.
- *This makes it for easy transport & assembly of SBM, which can be employed in remote places.
- *Carboniferous rock is probably most suitable environment for SBM.

SBM, Shaft Boring Machine IV

*Formation which has a large amount of water in flow must be sealed by grout before SBM begins boring.

SBM Shaft Boring Machine



Reference I

*SME Mining Engineering Handbook, 2nd Edition,
Volume 2, Page 1621 to 1631

Reference I

*SME Mining Engineering Handbook, 2nd Edition,
Volume 2, Page 1621 to 1631

Reference Others

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final size of the hole and the number of reaming stages. Structural analysis is required and is usually performed by computer, taking into account necessary balancing of the tool, achieved by optimum positioning of the cutters (Fig. 17.4.64). In the past, the primary body of the reamer was often supplemented with stages mounted below, resulting in a "Christmas-tree"-shaped tool (Fig. 17.4.65). Because of large torque requirements, this system is presently used only for shaft-sized holes, together with large non-rotating stabilizers. In another configuration, extensions are attached to the primary body. By doing this, an increase of the final diameter is achieved by adding more cutters in the same planes as the cutters on the primary body itself.

Gearred reamers are being used for reduction of required torque when compared to conventional reamers. The bit turns at a slower rate than the drill pipe due to the gearing mechanism. Shaft reamers are used for very large-diameter holes. This is a separate machine that is installed after the standard-sized raise has been completed, and is used for enlarging the raise to the full diameter required. The shaft reamer stabilizes itself by side jacks pressing against the hole and develops both drilling load and torque reacting against the anchored portion of the machine. The cuttings fall freely down the previously completed raise. Shaft lining can be installed when drilling. A service hoist to the working level is, however, necessary for men and materials.

The accepted measure of drilling performance is the rate of penetration, which depends primarily on bit design. The cutting action has to match the rock properties to optimize drilling rate (Fig. 17.4.66). During the reaming operation, in addition to the cutters, the design of the reaming head also influences the raise-boring performance. If no other source of information is available, an analysis of the wear of presently used cutters can help in the design of an improved cutting structure. Independent variables when drilling are rock properties, which are usually characterized by compressive, tensile (indirect), and shear strength measurements. The drilling tool should break rock primarily in tension and shear, because the strength of rock in these modes of loading is minimal. Strength alone does not determine drilling performance; additional factors intervening are rock ductility and abrasivity. For example, strong but brittle rock can have better drilling response than weak, clayish rock. Abrasivity is dependent on the mineral content of rock and is responsible for the wear of drilling tool elements in contact with the rock face and cuttings. The properties of the rock mass, such as joints, faults, and structural features, also influence drill response and, therefore, must be evaluated. The lowest drilling cost will occur at the point where both cutter life and rate of penetration are optimized.

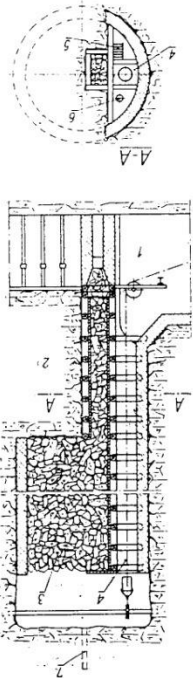
One of the most important technological aspects of boring is the removal of cuttings. Research has shown that a layer of cuttings on the uncut rock reduces penetration up to 40%. In other words, the energy of drilling is spent on regrounding the cuttings rather than penetration of the intact rock. In raise boring, this situation is encountered during drilling of the pilot hole. Standard drilling technology with compressed air, air and foam, or drilling fluid is used for cuttings removal. During the reaming operation, cuttings simply fall down the hole. Drilling load should be at the level required by the cutters to penetrate the rock and create fractures. Knowing the geometry of the cutting tool, it is possible to calculate the load exerted by particular cutters. Once the threshold strength of rock is reached, a further increase in drilling load does result in greater depth of penetration, but not in direct proportion to the increase of load. The general expression for this relationship is

$$D = W^a$$

(17.4.30)

dimension close to the hole diameter. These stabilizers are used for reaming up. Other types, for pilot holes and reaming down, have more clearance, allowing cuttings to pass. Once the pilot hole penetrates to the mine opening, the pilot bit and stabilizer are removed. After necessary servicing, the stabilizer is mounted on the drill stem and the reamer is put in the place of the small pilot-hole bit. Starting the hole (collaring) is a critical operation, and damage to the cutters or bit body can occur due to the uneven surface. The tension on the stem has to be adjusted to allow starting the hole without a highly variable, dynamic load acting on the drilling components. The stem is the longest at this stage of drilling and, therefore, prone to maximum elongation and twist. The design of the reamer depends on the

Fig. 17.4.59. Raising method with full diameter, short cuts, and immediate permanent lining.



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CONSTRUCTION OF DEVELOPMENT OPENINGS

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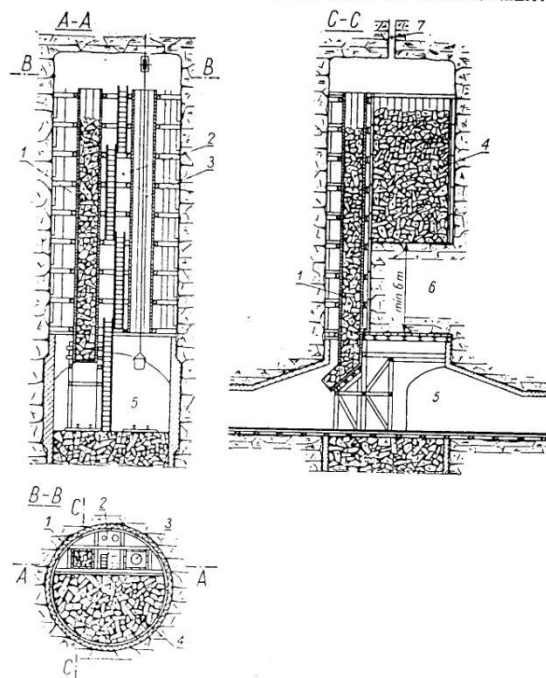


Fig. 17.4.58. Raising method with full diameter of the shaft, with temporary and a later-erected permanent lining.

1. chute compartment
2. ladder compartment
3. bucket compartment
4. storage of muck
5. lower shaft station
6. protective shelf
7. ventilation hole (to sump of existing upper shaft).

Conversion factor: 1 ft = 0.3048 m.

The disadvantages of a continuous boring system include a general requirement that rock conditions be similar within the length of the raise bore. Another disadvantage is that drilling in very hard rock is slow, and costs rapidly increase. Also access to upper and lower levels is necessary, or in other words, raise boring requires a pre-existing level. A raise-bored rock mass has to be dry. The initial cost of the rig is also high.

EQUIPMENT AND SYSTEM DESCRIPTION. Raise boring equipment is produced in the United States, in other industrialized western countries, and in the USSR. Machines that are designed primarily for soft rock are applied mainly in European coal fields. US manufacturers produce machines for hard- and soft-rock conditions as well.

There is a large variety of machines available that are designed for certain sets of rock conditions and nominal sizes and lengths of the finished hole. Typical characteristics of the machine specify the pilot-hole diameter, drill pipe diameter, reaming torque, type of drive (ac, dc, or hydraulic), its rpm and power (hp or kW), and type of gear reducer with corresponding rpms. Additional specifications are the pilot-hole thrust and feed rates, reaming pull, size of drill pipe, base plate and derrick dimensions with characteristics (especially important for underground application), and finally the type of transporter/erector, if applicable, and the water and air consumption.

Raise boring machines (RBMs) operate on the principle of first drilling a small pilot hole and then reaming the hole, in one or more stages, to the desired size. The modes of operation are shown in Fig. 17.4.61. The most frequently used is a conventional scheme with the pilot hole down and reaming upward. However, in some mining systems, box-hole drilling upwards in one stage is chosen.

The RBM consists of several parts (Fig. 17.4.62). The rig itself is composed of a rigid plate and a structure on which are assembled drilling, hydraulic, and electrical equipment. The rig has its own crawler driven by compressed air, or it is mounted on rail wheels for haulage. The rig can be disassembled and hauled in underground service tire vehicles. A pilot hole is drilled through the stem with stabilizers and a conventional drilling bit (Fig. 17.4.63). Drill rods have outer/inner diameters in a range of 8 to 12-7/8 in. (20.3 to 31.4 mm) to 4-3/4 to 5-7/16 in. (12.06 to 13.1 mm). The length of the sections for underground application is usually 5 ft (1.5 m), dictated by clearance restrictions. Connections are made with tapered thread joints that have to be kept absolutely clean.

The purpose of the stabilizer is to assure directional accuracy of the pilot hole. Several types are in common use. They are of a larger diameter than drill rods, with strips of tungsten carbide welded vertically or spirally on the outside, bringing the outer

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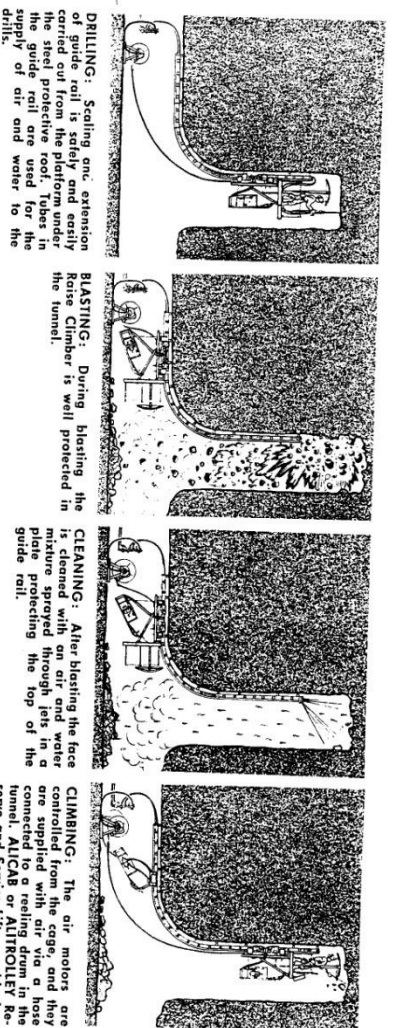


Fig. 17.4.60. Stages of operation of the Alimak mechanical raise system (courtesy: Alimak Co.).

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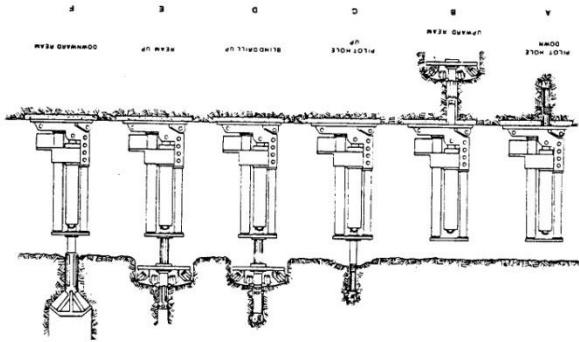


Fig. 17.4.61. Modes of operation in raise boring.

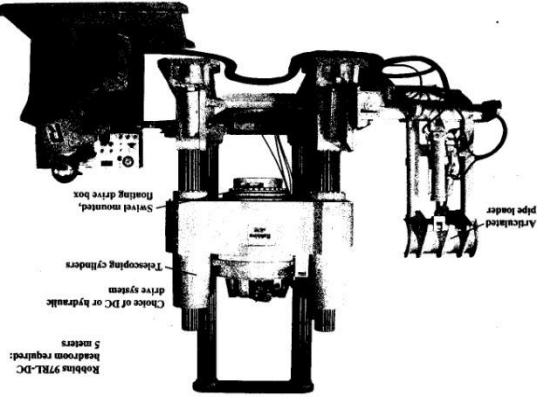


Fig. 17.4.62. Robbins 83RM-DC raise drill (courtesy: Robbins Co.).

where D is depth of penetration, H is the applied load, and d is a load exponent, which varies from 1.5 in soft rock to 1.0 in harder rocks.

During the reaming operation, the above equation is used for determination of load. The operator measures the rate of penetration over several minutes. He then adjusts the load and through several trials, finds the optimum load, above which the rate of penetration does not noticeably increase.

Rotary speed also affects the penetration rate. The higher the rotary speed, the faster the rate of penetration. However, restrictions other than the cutter-rock contact exist, namely, the wear of bearings and the tool itself. Vibration effects of uneven work by cutters result in wear of the drill pipe by rubbing against the wall of the pilot hole.

Table 17.4.12 lists the main parameters from several raise-boring projects.

METHOD SELECTION. The decision whether to drill raises from the top or bottom usually determines the access point. If only one end of the raise has adequate room, the machine must be put there. The extent to which raise boring will interfere with production must be evaluated. Location of the borer where less disturbance will be caused is preferred. Cost is also of major concern. Generally, drilling down and reaming up is cheaper than other configurations. Rock quality and its predicted response to drilling can influence the choice of direction as well.

LINING OR RAISES. In strong rocks without fractures, usually no support is needed. If support is required, shotcrete, rock bolts, with or without wire mesh, or steel lining can be applied. Shotcrete can be placed remotely or by workers from a cage. Rock bolts must be installed manually. Steel lining, required in unstable formations, is installed remotely, directly behind the reamer.

CONSTRUCTION OF DEVELOPMENT OPENINGS

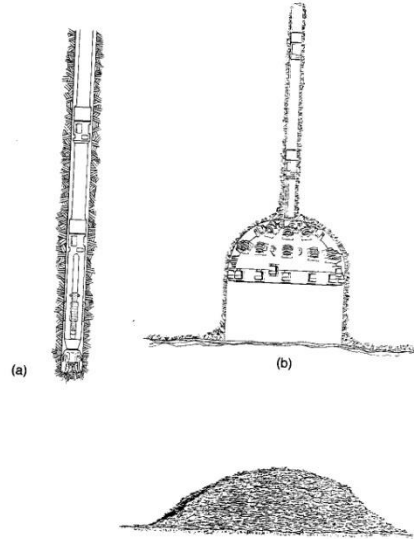


Fig. 17.4.63. Pilot hole (a) drilling and (b) reaming (courtesy: Robbins Co.).

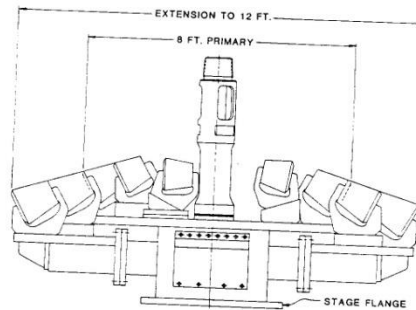


Fig. 17.4.64. Raise bit with extensions. Conversion factor:
1 ft = 0.3048 m.

For further discussion of rock boring, see Chapters 9.1 and 22.1.

17.4.2.6 Shaft Drilling and Boring

In recent years, a substantial effort has been made to improve methods and equipment used for drilling and for mechanical



Fig. 17.4.65. Top section of 16-ft (4.8-m) diameter reamer bar through the surface after 1490 ft (453 m) of reaming.

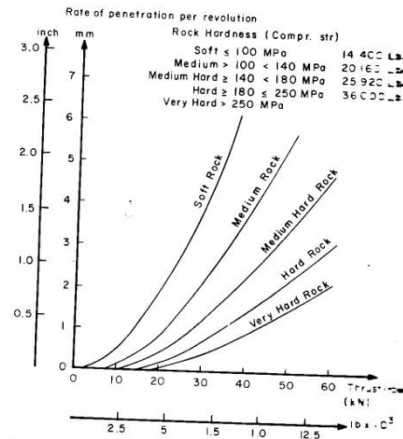


Fig. 17.4.66. Relationship between the rate of penetration thrust per revolution and rock strength.

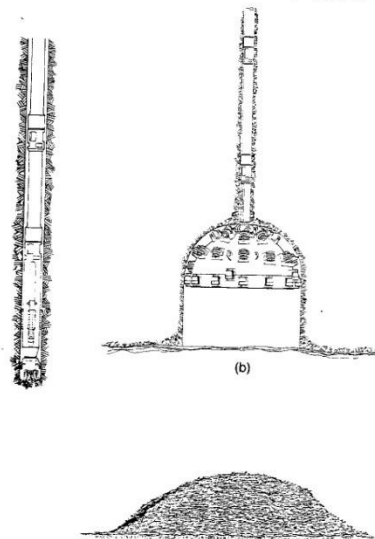


Fig. 17.4.63. Pilot hole (a) drilling and (b) reaming (courtesy: Robbins Co.).

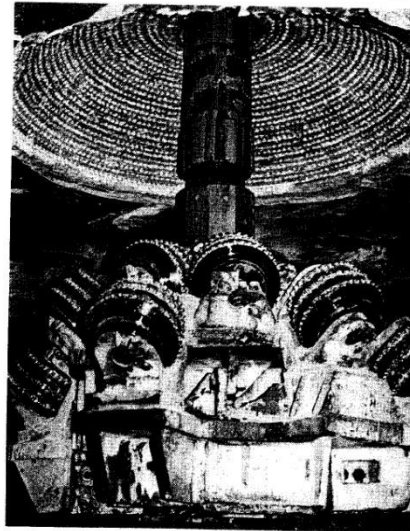


Fig. 17.4.65. Top section of 16-ft (4.8-m) diameter reamer breaking through the surface after 1490 ft (453 m) of reaming.

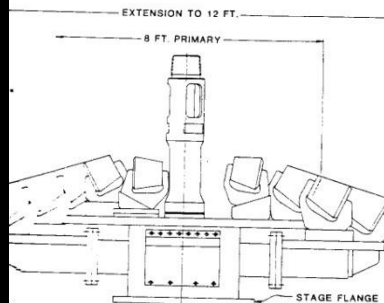


Fig. 17.4.64. Raise bit with extensions. Conversion factor: 1 ft = 0.3048 m.

for further discussion of rock boring, see Chapters 9.1 and

2.6 Shaft Drilling and Boring

In recent years, a substantial effort has been made to improve tools and equipment used for drilling and for mechanical

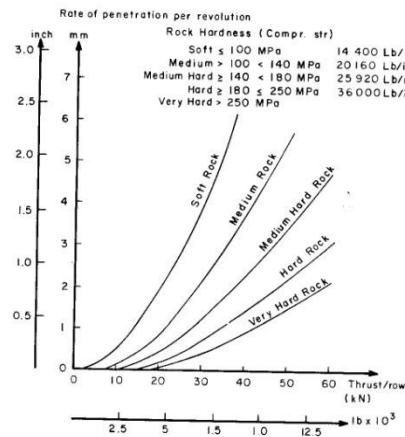


Fig. 17.4.66. Relationship between the rate of penetration thrust per row and rock strength.

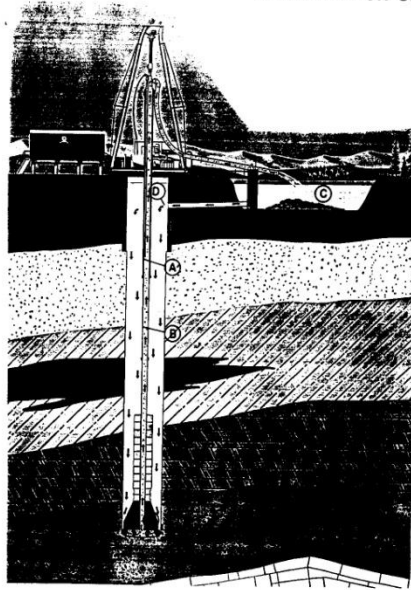


Fig. 17.4.68. Blind hole drilling system.

5. Organization of the drilling site, location of ponds for drilling fluid and a site for the fluid-making plant, and storage of lining components and materials.

DRILLING FLUID. The drilling fluid is a mixture of water and natural clay partly dispersed and partly colloidal, which by its hydrostatic head, keeps the wall of the hole in an equilibrium state and also transports chips of drilled rock and cools the bits. Ordinary drilling fluid should have a specific gravity of 1.15 to 1.25. Drilling fluid is prepared in mixers from which it flows to a tank and is pumped by slurry pumps to the shaft. During drilling, the shaft has to be filled completely with drilling fluid. To impose movement of the fluid, a compressed air airlift is used (Fig. 17.4.69). In small holes where the annular space between the drill string and hole wall is small, normal (right) circulation is applied. Air is injected in this space, and the drill fluid flows to the bottom through the drilling pipe, rising around it to lift cuttings to the surface. For large holes with a large sectional area of annular space between the drilling string and wall, the reverse system is used. Air-assisted, so-called reverse (or left) circulation systems, can be applied as single-, double-, and triple-wall pipe. In the simplest case, for example, a 3-in. (75-mm) pipe (if 13 3/8-in., or 33.4-mm, drill pipe is used) is run down the drill string, approximately 330 ft (100 m), sometimes to the end of the string. Compressed air is injected through the 3-in. (75-mm) pipe. When it rises in the annulus between the inner and drill pipes, it creates a mixture of air-drilling fluid of decreased

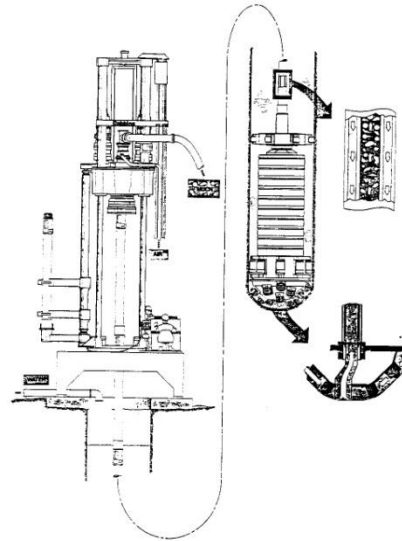
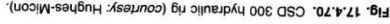


Fig. 17.4.69. Blind hole drilling rig (courtesy: Robbins Co.).

density. The positive pressure of the drilling fluid (with unchanged density) in the shaft induces circulation with a resulting high flow velocity in the drill pipe, enabling transport of cuttings to the surface. The more viscous the fluid, the less the fluid velocity need be. Usually, the maximum particle load any system can carry is 5% of its volume.

A new development is termed pumped reverse circulation. Instead of air, a submersible pump is installed as a part of the drilling string. This allows an increase in the volume of circulating fluid (in drill pipe of 12.8-in. or 300-mm bore, 4000 to 5300 gpm, or 15 to 20 m³/min) and consequently, better evacuation of chips from the face. To increase the size of cuttings transported, in addition to enlarging the flow volume, flow velocity can also be increased by restricting the cross-sectional area with skirts and shrouds at the very bottom of the hole. Here rock chips are lying exposed to unnecessary regrinding by the bit. Changing the angle of the sucking hole and adding chemicals to the drill fluid can also serve that purpose.

DRILLING EQUIPMENT. Originally, the drilling rigs used for large-hole drilling were converted heavy oil rigs. These were adequate when dealing with holes in a range of 10 to 12 ft (3 to 4 m), but for shafts of 15- to 25-ft (5- to 8-m) diameter, a rig had to be designed on a different set of principles. The rig must be able to produce large torque, and the required hoisting capacity must be very large. For example, CSD-300 is a rig manufactured by Hughes Micon, with six hydraulic motors producing 500,000 lb-ft (70 kNm) of torque. The power swivel is raised and lowered on hydraulic cylinders with a 2-million-lb (880-t) capacity (Fig. 17.4.70).



rotaries. They are appropriate for softer rocks because they do not damage the shaft wall formed by weak material. Stabilizers assure vertical alignment of the hole by restricting lateral movement of the drilling bit. By using multiple stabilization, it is possible to drill shafts with an alignment of 1 in 1000. Cutting cutters attached to the bit body. Great effort has been made over the last few decades to increase the reliability and life of cutters while permitting increased loading to be applied. Milled steel-tooth cutters are most suitable for soft rock, producing large cuttings. This gives the potential for rapid rates of penetration, but the performance of the cutters diminishes rapidly as the rock becomes dull. Tungsten carbide cutters are more appropriate for harder rock or in particularly abrasive formations; they require substantial life, but their rate of penetration is slow, although performance remains steady throughout the life of the cutter. Disk or scum cutters are not normally used for blind hole drilling because they tend to produce large chips that cannot be transported across the bottom of the hole and up to the surface. In soft and plastic strata, they can produce concentric furrows in the base of the bore hole with no excavation progress. The cutters themselves are mounted in saddles that are attached to a fabricated steel body of the bit. Various basic shapes have been used, including flat-bottomed, conical, hemispherical or a combination of shapes. The most advantageous, because very determining which shape is the best, is not enough evidence to determine conditions in particular projects make such comparisons difficult. Some believe that flat-bottom bits drill straighter holes than do conical bits; however, the latter give a better rate of penetration in medium to soft formations. Lining for Drilled Shafts. Types of permanent lining for drilled shafts depend on the character of rock formations intersected by the shaft. In stronger rocks, minimum lining requirements exist; therefore, shotcrete installation is sufficient. In fractured zones, a combination of rock bolts with mesh and shotcrete can be applied. For shafts with hosteling installations, usually a minimum thickness of 8 in. (200 mm) of concrete lining is required, needed for the fastening of consoles for shaft beams. These types of lining must be installed from a hanging stage using conventional technology. Permanent lining for steel shafts can be made of reinforced concrete or steel, as a single (with stiffening ribs) or double cylinder with concrete fill between the outer and inner segments. The work is performed at the surface and the lining is lowered to the shaft filled by drilling fluid. In the floating method, a steel bottom is made (welded) beside the shafts, lined with 16 to 23 in. (400 to 800 mm) of concrete, and set against the shaft. On this bottom, a first segment of lining 8 to 11.5 ft (2.5 to 3.5 m) high is welded. Extending the lining column with consecutive segments causes the weight of the lining to increase and, consequently, it sinks. Drilling fluid is displaced by lining with the equivalent volume. When the lining has positive buoyancy, by pouring water into it, the required balance can be achieved. The rate of lining placement is about 30 ft/day (10 m/day). The rate of lining placement in the shaft and the formation pressure, a double steel-cylinder lining with concrete filling the space between them can be made by either sinking it into the shaft, using steel segments with the required curvature and an automatic welding device. The concrete inside the lining is usually placed vertically with uniform spacing method. Injectors are placed vertically with uniform spacing. The space is then filled with gravel, sand, and aggregate (and grout in the desired proportion) and cement of high quality is pumped.

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